

The 25 questions most likely to win you marks tomorrow

These are the **highest-probability questions** for the Kerala Plus Two Chemistry SAY exam, each with a **full answer in simple English** that you can learn and write almost word-for-word. They cover roughly **40+ of the 60 theory marks** if they appear. The  **repeats yearly** ones come back almost every year.

Honest note: I can't promise the exact paper (no one can). But learn these 25 well and you are set up to pass and score high.

How to use this in your last hours

1. **Read each answer once, slowly.** Understand it, don't just stare.
2. Tap 🧠 **Cover answers** at the top. Now read the question, say the answer in your head, then tap/hold the card to check.
3. For questions with 🖋️ **Draw this**, practise the drawing on paper once.
4. Do all 25 once. Then do only the ones you got wrong again.
5. **In the exam:** always write the equation/formula line and balance it — you get marks for steps.

Q1

4 marks

 repeats yearly

Aldehydes/Ketones

Distinguish: (a) ethanal and propanone (b) propanal and propanone.

ANSWER (LEARN THIS)

(a) Ethanal vs propanone: Both give a positive iodoform test, so iodoform cannot separate them. Use **Tollens' reagent** (ammoniacal silver nitrate). Warm each with it. Ethanal is an aldehyde, so it gives a shiny **silver mirror**. Propanone is a ketone and gives no silver mirror. So the one giving the silver mirror is ethanal.

(b) Propanal vs propanone: Use **Fehling's solution** (blue). Warm each with it. Propanal is an aldehyde, so it gives a **red-brown precipitate** (Cu_2O). Propanone is a ketone, so it stays blue with no precipitate. So the one giving the red-brown precipitate is propanal.

Q2

4 marks

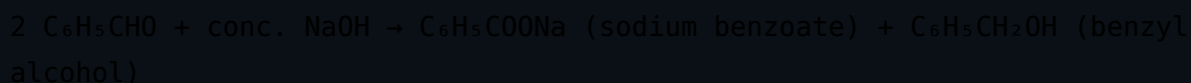
 repeats yearly

Aldehydes/Ketones

What is the Cannizzaro reaction? Give it for benzaldehyde and explain why benzaldehyde gives it but acetaldehyde does not.

ANSWER (LEARN THIS)

The **Cannizzaro reaction** happens with an aldehyde that has **no alpha-hydrogen**. When such an aldehyde is heated with strong concentrated alkali (NaOH), two molecules react: one is **oxidised** to the acid (its sodium salt) and the other is **reduced** to an alcohol. This is called disproportionation.



Benzaldehyde has **no alpha-hydrogen**, so it cannot do aldol condensation and instead does Cannizzaro. Acetaldehyde (CH_3CHO) **has alpha-hydrogens**, so it prefers aldol condensation and does not give the Cannizzaro reaction.

Q3

3 marks

Aldehydes/Ketones

Why are aldehydes more reactive than ketones in nucleophilic addition?

ANSWER (LEARN THIS)

1. Electronic reason: A ketone has two alkyl groups that push electrons towards the carbonyl carbon and lower its positive charge. An aldehyde has only one such group, so its carbon stays more positive and attracts the nucleophile more strongly.

2. Steric reason: The two bulky groups in a ketone block the nucleophile. An aldehyde has only one group, so the nucleophile reaches the carbon easily.

Because of both reasons, aldehydes react faster than ketones.

Q4

3 marks

 repeats yearly

Aldehydes/Ketones

Which compounds give a positive iodoform test?

ANSWER (LEARN THIS)

A yellow precipitate of **iodoform** (CHI_3) forms with compounds that have a $\text{CH}_3\text{CO}-$ group or a $\text{CH}_3\text{CH}(\text{OH})-$ group (alkali + iodine first turn the $\text{CH}_3\text{CH}(\text{OH})-$ group into $\text{CH}_3\text{CO}-$):

- Ethanal (CH_3CHO) · Methyl ketones like propanone (CH_3COCH_3) · Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) · Secondary alcohols of type $\text{CH}_3\text{CH}(\text{OH})\text{R}$.

So a yellow iodoform precipitate shows the compound has a $\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$ group.

Q5

3 marks

 repeats yearly

Alcohols/Phenols

Why is phenol more acidic than ethanol? How do nitro groups change it?

ANSWER (LEARN THIS)

When phenol loses H^+ it forms the **phenoxide ion**. The negative charge spreads into the ring by resonance, so the ion is **stable**. A stable ion forms easily, so phenol gives up H^+ easily and is acidic.

When ethanol loses H^+ , the charge stays on the oxygen and the ethyl group pushes electrons onto it, making it **less stable**. So ethanol is a much weaker acid.

Nitro groups ($-\text{NO}_2$) pull electrons away. At the ortho/para positions they spread the charge more and make phenoxide more stable, so nitrophenols are more acidic. 2,4,6-trinitrophenol (picric acid) is a strong acid.

Q6

3 marks

 repeats yearly

Alcohols/Phenols

Write the Reimer-Tiemann reaction.

ANSWER (LEARN THIS)

When phenol is heated with **chloroform** (CHCl_3) and aqueous **KOH** ($\sim 340\text{ K}$) and then treated with acid, a $-\text{CHO}$ group enters at the **ortho** position, giving **salicylaldehyde** (2-hydroxybenzaldehyde).



Draw this: a benzene ring with $-\text{OH}$ on one carbon and $-\text{CHO}$ on the next-door (ortho) carbon.

Q7

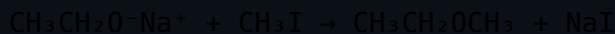
3 marks

Alcohols/Phenols/Ethers

What is Williamson's ether synthesis? Why use a primary alkyl halide?

ANSWER (LEARN THIS)

An **alkoxide** ($\text{R-O}^-\text{Na}^+$) reacts with an **alkyl halide** ($\text{R}'\text{-X}$) to give an ether ($\text{R-O-R}'$). It works by **SN2**.



We use a **primary** halide because the alkoxide is also a strong base. With secondary or tertiary halides the base causes **elimination** (an alkene) instead of the ether. A primary halide has little crowding, so substitution wins and the ether forms in good yield.

Q8

3 marks

Alcohols/Phenols

How do you tell apart 1°, 2°, 3° alcohols (Lucas test)? Test for phenol?

ANSWER (LEARN THIS)

Lucas test (conc. HCl + anhydrous ZnCl_2), watch for cloudiness at room temperature:

• **Tertiary (3°)**: turbidity at once. • **Secondary (2°)**: turbidity in about 5 minutes. • **Primary (1°)**: no turbidity in the cold (needs heating).

Test for phenol: add neutral $\text{FeCl}_3 \rightarrow$ phenol gives a **violet (purple) colour**; alcohols do not.

Q9

3 marks

 repeats yearly

Amines

Arrange aniline, ammonia, methylamine in increasing order of basic strength. Explain.

ANSWER (LEARN THIS)

Order: aniline < ammonia < methylamine.

- **Aniline** is weakest: its nitrogen lone pair is pulled into the ring by resonance, so it is less free to accept H^+ .
- **Methylamine** is strongest: the methyl group pushes electrons (+I) onto nitrogen, so the lone pair is more available and the ion formed is more stable.
- **Ammonia** has no ring and no pushing group, so it is in between.

Q10

3 marks

 repeats yearly

Amines

Give the importance of diazonium salts with two reactions.

ANSWER (LEARN THIS)

Aryl diazonium salts (Ar-N_2^+) are useful because the $-\text{N}_2^+$ group can be replaced by many groups that are hard to attach to a ring directly, so we can make many substituted benzenes.

Sandmeyer: $\text{Ar-N}_2^+\text{Cl}^- + \text{CuCl} \rightarrow \text{Ar-Cl} + \text{N}_2$ ($\text{CuBr} \rightarrow \text{Ar-Br}$, $\text{CuCN} \rightarrow \text{Ar-CN}$)

Coupling: $\text{Ar-N}_2^+ + \text{phenol} \rightarrow \text{p-hydroxyazobenzene}$ (an azo dye)

These reactions are very important in making dyes.

Q11

3 marks

Amines

Describe the carbylamine test and the Hoffmann bromamide reaction.

ANSWER (LEARN THIS)

Carbylamine test: heat a **primary amine** with chloroform and alcoholic KOH $\rightarrow$ a foul-smelling isocyanide forms. Only 1° amines give this smell, so it detects primary amines.

$\text{R-NH}_2 + \text{CHCl}_3 + 3\text{KOH} \rightarrow \text{R-NC} + 3\text{KCl} + 3\text{H}_2\text{O}$

Hoffmann bromamide: an amide with Br_2 and strong NaOH gives a primary amine with **one carbon less**.

$\text{R-CONH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{R-NH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$

Q12

3 marks

 repeats yearly

Haloalkanes

Compare SN1 and SN2 reactions.

ANSWER (LEARN THIS)

SN1 : two steps (forms a carbocation first); rate = $k[\text{RX}]$ (depends only on substrate); gives racemisation (both mirror images); favoured by **tertiary** halides.

SN2 : one step; rate = $k[\text{RX}][\text{nucleophile}]$ (depends on both); gives inversion (Walden inversion); favoured by **primary** halides (less crowding).

Q13

3 marks

Haloarenes

Why is chlorobenzene less reactive than chloromethane towards substitution?

ANSWER (LEARN THIS)

In chlorobenzene the chlorine lone pair goes into the ring by resonance, giving the C–Cl bond **partial double-bond character** — it becomes short and strong and hard to break. The carbon is sp^2 (more electronegative) and holds the chlorine tightly, and the electron-rich ring repels the nucleophile. So substitution is difficult. In chloromethane the carbon is sp^3 with a normal single C–Cl bond, so substitution is easy.

Q14

3 marks

 repeats yearly

Coordination

Using CFT, explain why $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic but $[\text{CoF}_6]^{3-}$ is paramagnetic.

ANSWER (LEARN THIS)

Both have Co^{3+} (d^6) and are octahedral. The d-orbitals split into a lower t_{2g} set (3) and an upper e_g set (2), with a gap Δ_0 .

- $[\text{Co}(\text{NH}_3)_6]^{3+}$: NH_3 is a strong-field ligand $\rightarrow \Delta_0$ large $\rightarrow$ all 6 electrons pair in t_{2g} ($t_{2g}^6 e_g^0$) $\rightarrow$ no unpaired electrons $\rightarrow$ **diamagnetic** (low-spin).
- $[\text{CoF}_6]^{3-}$: F^- is a weak-field ligand $\rightarrow \Delta_0$ small $\rightarrow$ electrons spread out ($t_{2g}^4 e_g^2$) $\rightarrow$ 4 unpaired electrons $\rightarrow$ **paramagnetic** (high-spin).



Draw this: the splitting diagram — three lower t_{2g} lines and two upper e_g lines with the gap Δ_0 .

Q15

3 marks

 repeats yearly

Coordination

Write the IUPAC names of (a) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (b) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (c) $[\text{Co}(\text{en})_3]\text{Cl}_3$.

ANSWER (LEARN THIS)

- (a) Potassium hexacyanidoferrate(II)
 (b) Hexaamminechromium(III) chloride
 (c) Tris(ethane-1,2-diamine)cobalt(III) chloride

Rules: name ligands first in alphabetical order, anionic ligands end in -o (cyanido), use bis/tris for numbers, oxidation state in Roman numerals, and a complex anion ends in -ate (ferrate).

Q16

3 marks

Coordination

Define a chelate and an ambidentate ligand with examples.

ANSWER (LEARN THIS)

Chelate: when a ligand with two or more donor atoms binds the same metal and makes a **ring**.

Example: ethylenediamine (en) binds through two N atoms and forms a five-membered ring.

Chelates make the complex extra stable.

Ambidentate ligand: a ligand with two different donor atoms that uses only one at a time.

Example: NO_2^- (binds through N or O); SCN^- (binds through S or N).

Q17

3 marks

 repeats yearly

d- and f-Block

What is lanthanoid contraction? Give two consequences.

ANSWER (LEARN THIS)

Lanthanoid contraction is the steady decrease in size of the lanthanoids from La to Lu. It happens because the 4f electrons shield the nucleus poorly, so the increasing nuclear charge pulls the electrons closer and the size shrinks.

Consequences: (1) Second and third transition series elements of a group become almost the same size (e.g. Zr and Hf), so they are hard to separate. (2) The basic strength of lanthanoid hydroxides decreases slowly across the series.

Q18

3 marks

d- and f-Block

Why do transition metals show variable oxidation states and why are their ions coloured?

ANSWER (LEARN THIS)

Variable oxidation states: the $(n-1)d$ and ns orbital energies are very close, so electrons can be lost from both. The metal can lose different numbers of electrons, giving many oxidation states (Fe is +2 and +3; Mn from +2 to +7).

Colour: their ions have partly filled d-orbitals. An electron absorbs visible light and jumps from a lower to a higher d-level (a **d-d transition**); the leftover colour is what we see. Ions with empty or full d-orbitals are usually colourless.

Q19

3 marks

repeats yearly

Electrochemistry

Calculate the EMF of $\text{Zn}|\text{Zn}^{2+}(0.1\text{ M})||\text{Cu}^{2+}(1.0\text{ M})|\text{Cu}$ at 298 K. $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76\text{ V}$, $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34\text{ V}$.

ANSWER (LEARN THIS)

Step 1 (E°_{cell}): $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34 - (-0.76) = \mathbf{1.10\text{ V}}$.

Step 2 (Q): $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, $n = 2$, $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 0.1/1.0 = 0.1$.

$$E = E^\circ_{\text{cell}} - (0.0591/n) \log Q = 1.10 - (0.0591/2) \log(0.1) = 1.10 + 0.0296 = \mathbf{1.13\text{ V}}$$

Answer: $E \approx \mathbf{1.13\text{ V}}$. (Always write the Nernst formula line first — step marks.)

Draw this: the galvanic cell — two beakers, an electrode in each, a salt bridge on top and a voltmeter; mark anode (−) and cathode (+).

Q20

3 marks

Electrochemistry

State Kohlrausch's law. Why does molar conductivity increase on dilution but conductivity decrease?

ANSWER (LEARN THIS)

Kohlrausch's law: at infinite dilution, the molar conductivity of an electrolyte equals the sum of the conductivities of its cation and anion. It is used to find Λ°_{m} of weak electrolytes and the degree of dissociation.

On dilution: conductivity (κ) is per unit volume; with fewer ions in each unit volume, κ decreases. Molar conductivity (Λ_{m}) counts all ions from one mole; on dilution more electrolyte dissociates and ions move freely, so Λ_{m} increases to a maximum.

Q21

3 marks

repeats yearly

Chemical Kinetics

A first-order reaction has $k = 1.15 \times 10^{-3}\text{ s}^{-1}$. How long will 5 g of reactant take to fall to 3 g?

ANSWER (LEARN THIS)

$$\text{First order: } k = (2.303/t) \log([R]_0/[R]) \Rightarrow t = (2.303/k) \log([R]_0/[R])$$

$$t = (2.303 / 1.15 \times 10^{-3}) \times \log(5/3) = 2002.6 \times 0.2219 = \mathbf{444\text{ s}} \text{ (about 7.4 minutes).}$$

Q22

3 marks

Chemical Kinetics

Difference between order and molecularity. What is a pseudo first-order reaction?

ANSWER (LEARN THIS)

Order: the sum of the powers of concentration terms in the experimental rate law; can be 0, fractional or whole; for the overall reaction.

Molecularity: the number of species in one elementary step; always a whole number; only for elementary reactions.

Pseudo first-order: a reaction that is really higher order but acts first-order because one reactant is in large excess (its concentration hardly changes). Example: acid hydrolysis of ethyl acetate (water in excess).

Q23

3 marks

Solutions

1.0 g of a non-electrolyte in 50 g of benzene lowers the freezing point by 0.40 K ($K_f = 5.12$). Find its molar mass. State Raoult's law.

ANSWER (LEARN THIS)

$$M = (K_f \times w_{\text{solute}} \times 1000) / (\Delta T_f \times w_{\text{solvent}}) = (5.12 \times 1.0 \times 1000) / (0.40 \times 50) = 256 \text{ g/mol}$$

Raoult's law: for a solution of volatile liquids, the partial vapour pressure of each component equals its pure vapour pressure times its mole fraction: $p_A = p_A^\circ \times x_A$.

Q24

3 marks

 repeats yearly

Biomolecules

Two differences between DNA and RNA. What is denaturation? Why is sucrose non-reducing?

ANSWER (LEARN THIS)

DNA vs RNA: (1) DNA has deoxyribose and thymine; RNA has ribose and uracil. (2) DNA is double-stranded and stores genetic information; RNA is usually single-stranded and helps make proteins.

Denaturation: loss of a protein's natural 3-D shape (secondary/tertiary structure) by heat or acid, while the amino-acid sequence stays the same; the protein loses its function (e.g. boiling an egg).

Sucrose non-reducing: the reducing groups of glucose and fructose are both locked in the glycosidic bond, so there is no free reducing group — it cannot reduce Tollens'/Fehling's.

Q25

3 marks

 repeats yearly

Everyday Life

Explain the cleansing action of soap. Difference between antiseptics and disinfectants, and between soaps and detergents.

ANSWER (LEARN THIS)

Cleansing action: a soap molecule has an oily tail (likes grease) and an ionic head (likes water). In water the tails go into the grease and the heads stay outside, forming a tiny ball called a **micelle** with the dirt trapped inside. The micelles float away with water, cleaning the cloth.

Antiseptic vs disinfectant: antiseptics go on living tissue (Dettol, dilute phenol); disinfectants go on non-living surfaces (concentrated phenol). The same chemical can be both, depending on concentration.

Soap vs detergent: soaps fail in hard water (form scum with $\text{Ca}^{2+}/\text{Mg}^{2+}$); detergents work in hard water because their Ca/Mg salts are soluble.



Draw this: a micelle — grease in the centre, soap tails pointing in, round heads pointing out into water.

You have everything you need. Stop reading new material now — learn these 25, draw the diagrams, and rest before the exam. You can do this. 